

Hana No Hana: Artist's Statement

Academic Medicine, Volume 84(2): 198. 2009

Stolz, Donna Beer PhD

Author Information

Dr. Stolz is associate professor, cell biology and physiology, and assistant director, Center for Biologic Imaging, University of Pittsburgh Medical School, Pittsburgh, Pennsylvania.

Editor's Note: This Teaching and Learning Moments essay was contributed as a companion to this month's AM Cover Art selection, which appears on the cover.

As assistant director of the Center for Biologic Imaging at the University of Pittsburgh, I am often asked by various clinicians to assist in projects using microscopy to augment their clinical research. B. J. Ferguson, MD, a surgeon in otolaryngology at the University of Pittsburgh Medical Center, invited me to collaborate on a project to identify bacterial biofilms in patients with chronic rhinosinusitis, especially those cases refractory to antibiotic treatment. I was given myriad sinus biopsies from a variety of patients and, after the usual time-consuming processing, I set about looking for bacteria and/or biofilms using transmission electron microscopy. During the many hours I spent at the electron microscope searching for the microbes and biofilm communities, I often came across spectacular scenery through the oculars. One such frame caught my eye and reminded me of a field of flowers. Sinus epithelium is very specialized and, in its normal state, has both microvilli and cilia decorating the apical plasma membrane surface. This specific section had a nice combination of cilia in cross section and microvilli in longitudinal format, resembling flowers and tall grass, respectively. After taking the frame, the black and white image was pseudocolored to represent the field of flowers I had envisioned, with the cross-sectioned cilia as flowers and the microvilli as grass, all against a blue sky.

My observation of the resemblance between the cilia and microvilli and flowers and grass has been influenced by, of all things, my study of the Japanese language. The Japanese words for flower and nose are homonyms: *hana*. However, the characters the Japanese use in writing these words are different. The characters written in the lower right-hand corner of the image read “nose flowers” to reflect that the depicted “field of flowers” was derived from a paranasal sinus mucosal biopsy. The final product was meant to loosely represent a Japanese watercolor or woodcut with a title written in Japanese on the print.